

# ARPON KAPURIA

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## EDUCATION

**National Institute of Technology Tiruchirappalli**  
Bachelor of Technology (*Computer Science & Engineering*)

**Tiruchirappalli, India**  
November 2020 – June 2024

**Kushtia Government College, Kushtia**  
HSC (*Science*) | GPA: 5 / 5

**Kushtia, Bangladesh**  
2019

**Kushtia Zilla College, Kushtia**  
SSC (*Science*) | GPA: 5 / 5

**Kushtia, Bangladesh**  
2017

## SKILLS

- **Programming Languages:** Python, C, C++, SQL, Bash
- **AI / ML:** PyTorch, Hugging Face Transformers, PEFT, TRL, Unsloth, LangChain, LangGraph, vLLM
- **Backend & Data:** FastAPI, Ray, PostgreSQL, Redis, Vector Databases
- **Infrastructure & MLOps:** MLflow, DVC, Docker, Kubernetes, Terraform, AWS, Git, CI/CD
- **Observability:** LangSmith, Prometheus, Grafana, OpenTelemetry
- **Spoken Languages:** Bengali (Native), English (IELTS 7.5), Hindi (Conversational), German (A1.1)

## EXPERIENCE

**Advanced Machine Intelligence Research Lab**  
Research Assistant | *Advisor: Prof. M. Firoz Mridha*

**Dhaka, Bangladesh**  
November 2025 – Present

- Engaged in research on Generative NLP and assisted in implementing deep learning models.

**Indian Institute of Technology Bombay**  
Research Intern | MeDAL Lab | *Advisor: Prof. Amit Sethi*

**Mumbai, India**  
May 2023 – July 2023

**Project 1:** Enhancing Self Supervised Learning framework - BYOL

- Conducted an in-depth survey on self-supervised learning frameworks and reproduced BYOL and sim-CLR in **PyTorch** to establish a baseline for improvements.
- Introduced changes to BYOL by incorporating a novel loss function, algorithmic modifications and architectural adjustments to improve representation learning.

**Project 2:** Radiology DICOM Image Anonymization

- Created a user-friendly **Flask** based system for DICOM image processing and anonymization using Pydicom, ensuring HIPAA-compliant privacy across **5,000+ medical scans**.
- Evaluated the CRAFT model for text detection in X-ray images and extracting clinically relevant annotations.

**National Institute of Technology Tiruchirappalli**  
Undergraduate R&D Assistant | Industrial Automation Lab

**Tiruchirappalli, India**  
October 2022 – February 2023

**Project:** Assam Smart Home Automation

*Advisors: Prof. M. Brindha, Prof. G. S. Ilango*

- Developed a **Flutter** app to automate energy operations and digital billing for a solar-powered micro-grid in Assam, India, serving **200+ households**.
- Integrated Flutter frontend with Django-based REST API and deployed an MQTT broker enabling real-time data exchange with **50+ IoT sensors** for energy consumption and control.
- Project supported by the Ministry of Science & Technology, Government of India, under the initiative **SUSTENANCE** to achieve Novel Carbon Neutral Energy Communities.

## OPEN SOURCE CONTRIBUTIONS

- Huggingface/[transformers](#)
- LangChain/[langchain](#)
- PyTorch/[pytorch](#)

## SELECTED PROJECTS

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### Agentic RAG System with Hybrid Retrieval & Scalable Deployment

March 2026 – April 2026

*Python, FastAPI, Ray, LangGraph, vLLM, Qdrant, Neo4j, PostgreSQL, Docker, Kubernetes, Terraform, AWS, Prometheus, Grafana, OpenTelemetry*

- Designed an end-to-end **Agentic RAG** system with a planner routing queries across **hybrid retrieval**, using **HyDE**, **query rewriting**, and a **cross-encoder reranker** to improve **multi-hop reasoning** beyond embedding-only pipelines.
- Deployed **LLM inference** with AWQ quantization, **continuous batching**, and **tensor parallelism**; served embeddings with **fractional GPU allocation** and compilation optimizations, scaling to **128 concurrent sequences per replica** under **autoscaled GPU worker groups**.
- Provisioned **multi-AZ infrastructure** and configured separate CPU and GPU provisioners with **30-second GPU scale-to-zero** and **spot consolidation**, reducing **idle compute cost** without compromising **autoscaling responsiveness**.
- Instrumented the **full request path** with traces, metrics, and dashboards; evaluated retrieval and generation quality via **RAGAS** and an **LLM-as-a-Judge pipeline** against a curated **golden dataset**.

### Cold Email Generator

April 2025

*Python, LangChain, FAISS, RAGAS, Llama-4, BeautifulSoup (BS4)*

- Automated an end-to-end **multimodal RAG system** to streamline graduate application outreach by aligning applicant profiles with professor research via web scraping, text parsing, and CV image analysis, achieving a **92.6% relevance score** (RAGAS).
- Designed a LangChain-based pipeline leveraging **LLaMA-4 Maverick (Groq AI)**, **Jina Embeddings v3**, and **FAISS**, leveraging multimodal embeddings to generate contextually personalized cold emails.
- Incorporated **Cohere Reranker v3.5**, achieving a **Precision@K of 87.3%**, with robustness validated across diverse input formats (PDFs, HTML, images).

### Malicious Website Detection Using Machine Learning

October 2022 – November 2022

*Python, Scikit-Learn, Flask, Docker, JavaScript, Chrome Extension tools*

- Implemented and deployed a machine-learning model via **Flask** and **Docker**, achieving **94% accuracy** for real-time malicious website detection.
- Trained and benchmarked **4 models** – KNN, SVM, LR & MLP – and optimized hyperparameters, resulting **14% enhancement** in model performance .
- Built a Chrome extension using JavaScript to extract **27 real-time features** (e.g., URL length, domain authority) from webpages and integrated user feedback collection, used by **100+ test users**.
- Applied a continual learning pipeline that retrains the model every **24 hours** on the feedback data, reducing false positives by **almost 6%** and increasing detection robustness.

## ACHIEVEMENTS

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- 2020**    **ICCR Scholarship** – Ministry of External Affairs, Government of India; awarded for academic excellence and funded my bachelor studies.
- 2017**    **Academic Excellence Award** – Chamber of Commerce, Kushtia; SSC Board (10th) Results.
- 2015**    **General Grade Scholarship** – Government of Bangladesh; JSC Board (8th) Results.
- 2012**    **1st Place** – Zilla Shilpakala Academy, Kushtia; Art Competition (Independence Day).
- 2011**    **2nd Place** – Bangladesh Udichi Shilpigoshthi, Kushtia; Art Competition (Bengali New Year 1418).

## REFERENCES

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**Dr. M. Brindha**, Associate Professor

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**Amit Sethi**, Professor

Department of Electrical Engineering, IIT Bombay, India | [asethi@iitb.ac.in](mailto:asethi@iitb.ac.in)